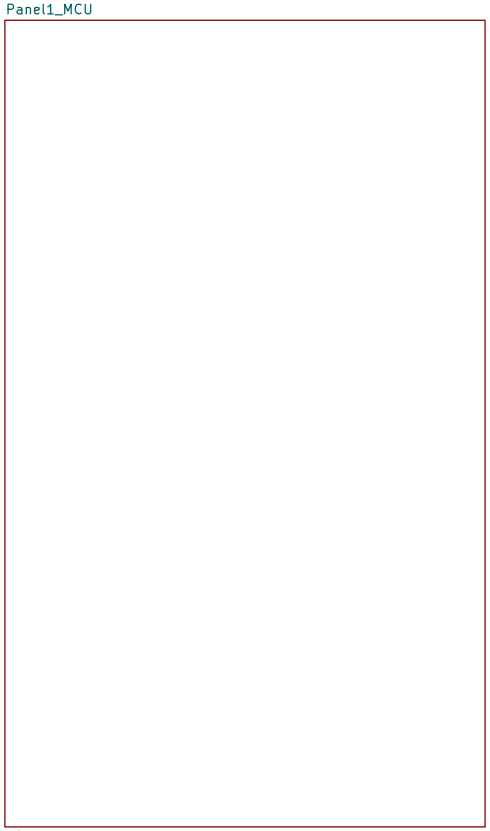
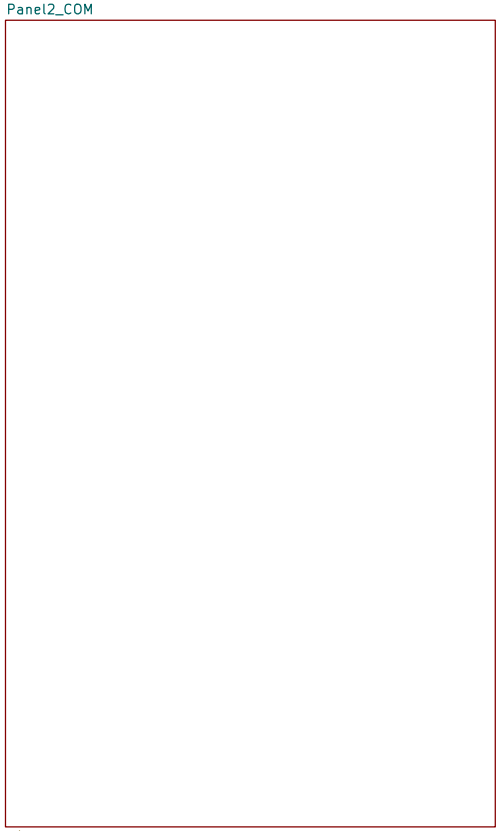
Panel1_MCU



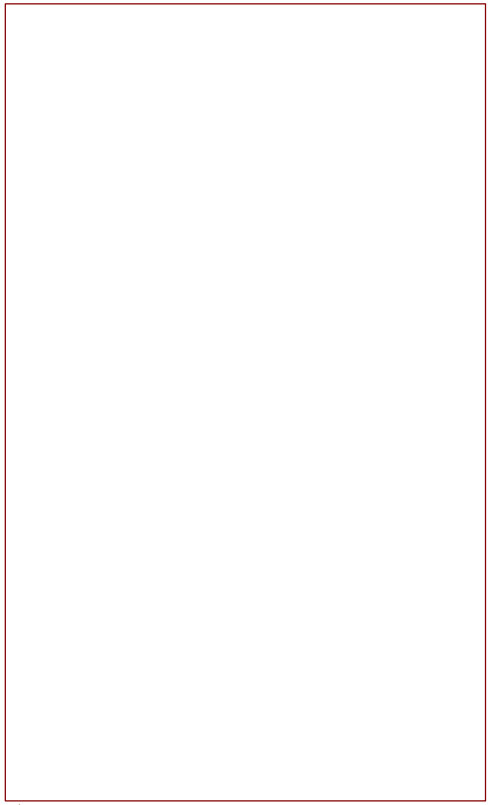
Fájl: Panel1_MCU.kicad_sch

Panel2_COM



Fájl: Panel2_COM.kicad_sch

Panel3_GLM-PSU



Fájl: Panel3_GLM-PSU.kicad_sch

Panel5_FLIR



Fájl: Panel5_FLIR.kicad_sch

Panel4_BAT_Plate



Fájl: Panel4_BAT_Plate.kicad_sch

python -m easyeda2kicad --full --lcsc_id C19268026 --output "D:\Data\T.E.E.M.O\Circuit\CanSat_V2\panel

Sheet: /
File: CanSat.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 1/22

Board Connector

CN1
61082-041402LF

3.3V	1	2	Sensors_3.3V		
Standby_3.3V	3	4	Standby_3.3V		
GND	5	6			GND
LP_SCK	7	8	LP_I2C_SCL		
LP_MOSI	9	10	LP_I2C_SDA		
LP_MISO	11	12	GLM_LP_CS		
GND	13	14	HARD_RESET		
Radio_SCK	15	16			GND
Radio_MOSI	17	18	2G4MHZ_CS		
Radio_MISO	19	20	900MHZ_CS		
RADIO_RST	21	22	400MHZ_CS		
SIM7080_PWRKEY	23	24	170MHZ_CS		
170MHZ_INT	25	26	170MHZ_BUSY		
400MHZ_INT	27	28	400MHZ_BUSY		
900MHZ_INT	29	30	900MHZ_BUSY		
2G4MHZ_INT	31	32	2G4MHZ_BUSY		
GND	33	34			GND
SIM7080_TXD	35	36	SIM7080_RXD		
RANGE_TXD	37	38	RANGE _RXD		
VTX_TXD	39	40	VTX_RXD		

FLIR Connector

J6
Conn_01x15_Pin

1	Lepton_PWR	
2	Lepton_PWR	
3		GND
4	LEPTON_PWR_EN	
5	LEPTON_ENABLE	
6	LEPTON_RST	
7	I2C0_SCL	
8	I2C0_SDA	
9		GND
10	LEPTON_SPI3_CS	
11	LEPTON_SPI3_SCK	
12	LEPTON_SPI3_MISO	
13	LEPTON_VSYNC	
14	LEPTON_MasterCLK	
15		GND

LP CS decoder

U45
TI SN74LVC138ARSVR

GND
C76 100nf
Sensors_3.3V

LP_CS_DEMUX_A	1	Y0	15 GLM_LP_CS
LP_CS_DEMUX_B	2	Y1	14 MICS_LP_CS
LP_CS_DEMUX_C	3	Y2	13 ICM_LP_CS
		Y3	12 G-FORCE_LP_CS
		Y4	11 BARO_LP_CS
		Y5	10
		Y6	9
		Y7	7 Sensors_3.3V

VCC 16, G2B 5, G2A 4, G1 6, GND 8
LEDD9 220ohm R57
GND

Radio CS decoder

U44
TI SN74LVC138ARSVR

GND
C75 100nf
Sensors_3.3V

RADIO_CS_DEMUX_A	1	Y0	15 2G4MHZ_CS
RADIO_CS_DEMUX_B	2	Y1	14 900MHZ_CS
RADIO_CS_DEMUX_C	3	Y2	13 400MHZ_CS
		Y3	12 170MHZ_CS
		Y4	11 SPI_UART_CS
		Y5	10
		Y6	9
		Y7	7 Sensors_3.3V

VCC 16, G2B 5, G2A 4, G1 6, GND 8
LEDD8 220ohm R56
GND

GPS

Sensors_3.3V
GPS_RXD
GPS_TXD

3.3V	
RX	GPS_RXD
TX	GPS_TXD

Fájl: GPS.kicad_sch

I2C_Sensors

Sensors_3.3V
I2C0_SCL
I2C0_SDA

Sensors_3.3V	
I2C0_SCL	I2C0_SCL
I2C0_SDA	I2C0_SDA

Fájl: I2C_Sensors.kicad_sch

SiPM Connector

J5
Conn_01x15_Pin

1	SIPM_3.3V	
2		
3	I2C1_SCL	GND
4	I2C1_SDA	
5	GAMMA_RST	
6	MUON_RST	
7	TCA_RST	
8	LP_SPI_CS	
9	LP_SPI_SCK	
10	LP_SPI_MOSI	
11	LP_SPI_MISO	
12		GND
13	GAMMA_INT	
14	MUON_INT	
15		GND

Lepton_SiPM_Power

LP_I2C_SCL
LP_I2C_SDA
Standby_3.3V
3.3V

SCL	Lepton_PWR
SDA	Lepton_PWR
Standby_3.3V	SiPM_3.3V
3.3V	SiPM_3.3V

Fájl: Lepton_SiPM_Power.kicad_sch

SPI_Sensors

Sensors_3.3V
MICS_LP_CS
ICM_LP_CS
G-FORCE_LP_CS
BARO_LP_CS
LP_MISO
LP_MOSI
LP_SCK
LEPTON_VSYNC
MICS_CONVST

3.3V	
MICS_LP_CS	MICS_LP_CS
ICM_LP_CS	ICM_LP_CS
G-FORCE_LP_CS	G-FORCE_LP_CS
BARO_LP_CS	BARO_LP_CS
MISO	LP_MISO
MOSI	LP_MOSI
SCK	LP_SCK
ICM_FSYNC	LEPTON_VSYNC
MICS_CONVST	MICS_CONVST

Fájl: SPI_Sensors.kicad_sch

U53
114-00841-68

Sensors_3.3V
R94 47k
R95 47k
SD_CMD
Sensors_3.3V
SD_CLK
SD_DATA0
Sensors_3.3V
47k R97

	1	DAT2	EH 14
	2	CD/DAT3	EH 13
	3	CMD	
	4	VDD	
	5	CLK	
	6	VSS	
	7	DAT0	
	8	DAT1	
	9	CD1	EH 12
	10	CD2	EH 11

GND GND GND

C6_Programmer

Standby_3.3V
+3.3V

C6_TX	C6_TX
C6_RX	C6_RX
RESET	ESP_RST
BOOT	C6_BOOT

Fájl: C6_Programmer.kicad_sch

P4_Dual_USB

Standby_3.3V	PROG_USB_D+	PROG_USB_D+
	PROG_USB_D-	PROG_USB_D-
	High_Speed_USB_D+	High_Speed_USB_D+
USB_ID	High_Speed_USB_D-	High_Speed_USB_D-

Fájl: P4_Dual_USB.kicad_sch

U6
ESP32-P4-Module

Standby_3.3V
ESP_RST 87
C6_RX 4
C6_TX 5
C6_BOOT 9

LP_SPI_CS
LP_SPI_SCK
LP_SPI_MOSI
LP_SPI_MISO
GAMMA_INT
MUON_INT
GAMMA_RST
MUON_RST

ESP_LDO_VO4	66	GPIO0	88 LP_I2C_SCL
ESP_3V3	85	GPIO1	16 LP_I2C_SDA
ESP_3V3	86	GPIO2	17 LP_MISO
VBAT	84	GPIO3	18 LP_MOSI
ESP_EN	87	GPIO4	19 LP_SCK
LNA_OUT	2	GPIO5	20 LP_CS_DEMUX_A
C6_U0RXD Debug	4	GPIO7	22 LP_CS_DEMUX_B
C6_U0TXD UART	5	GPIO8	23 LP_CS_DEMUX_C
C6_IO4	14	GPIO9	24 MICS_CONVST
C6_IO5	13	GPIO10	25
C6_IO6	12	GPIO11	26
C6_IO7	11	GPIO12	27
C6_IO8 Enable	10	GPIO13	28
C6_IO9 BOOT_Mode	9	GPIO6	21
C6_IO13	8	GPIO20	29
C6_IO14	7	GPIO21	30
C6_IO15	6	GPIO22	31
		GPIO23	32
		GPIO24	50 PROG_USB_D-
		GPIO25	51 PROG_USB_D+
CSI_DATAN0	41	GPIO26	53 170MHZ_BUSY
CSI_DATAP0	42	GPIO27	54 400MHZ_BUSY
CSI_DATAN1 Camera MIPI	45	GPIO28	55 900MHZ_BUSY
CSI_DATAP1 100 Ω Diff	46	GPIO29	56 2G4MHZ_BUSY
CSI_CLKN	44	GPIO30	57 2G4MHZ_INT
CSI_CLKP	43	GPIO31	58 900MHZ_INT
		GPIO32	59 400MHZ_INT
		GPIO33	60 170MHZ_INT
		GPIO34	61
		GPIO35	62 Radio_MISO
		GPIO36	63 Radio_MOSI
		GPIO37	64 Radio_SCK
		GPIO38	65
DSI_DATAN0	39	GPIO39	67 I2C0_SCL
DSI_DATAP0	38	GPIO40	68 I2C0_SDA
DSI_DATAN1 Display MIPI	35	GPIO41	69 I2C1_SCL
DSI_DATAP1 100 Ω Diff	34	GPIO42	70 I2C1_SDA
DSI_CLKN	36	GPIO43	71
DSI_CLKP	37	GPIO44	72
		GPIO45	73 LEPTON_SPI3_CS
		GPIO46	74 LEPTON_SPI3_SCK
		GPIO47	75 LEPTON_SPI3_MISO
		GPIO48	76
		GPIO49	77 LEPTON_VSYNC
		GPIO50	78 LEPTON_MasterCLK
		GPIO51	79
		GPIO52	80 SD_CMD
		GPIO53	81 SD_CLK
		GPIO54	82 SD_DATA0
USB_DM Hi-Speed USB	48		RADIO_CS_DEMUX_A
USB_DP 480 Mib/s	49		RADIO_CS_DEMUX_B
			RADIO_CS_DEMUX_C

Low Power Core Dedicated GPOs
TOUCH CH: 0–11
ESP32-C6 integrated CO-processor
SDIO_D0-4 -> GPIO14-17
SDIO_CMD -> GPIO18
SDIO_CLK -> GPIO19
C6_INT
ADC1 CH: 4–7
JTAG USB D– JTAG USB D+
BOOT_Mode BOOT_Enable
Voltage reference by ESP_LDO_V04 Default is 1.8V
ADC2 CH: 0–5
GND GND GND GND GND GND GND GND
1 3 15 35 47 52 83
GND

SPI_to_UART

Sensors_3.3V
SPI_UART_INT
SPI_UART_CS
Radio_MISO
Radio_MOSI
Radio_SCK

3.3V			
INT			
CS			
MISO			
MOSI			
SCK			
GPIO0_U0_CTS		GPIO2_U1_CTS	
GPIO19_U0_DCD		GPIO23_U1_DCD	
GPIO20_U0_DTR		GPIO8_U1_DTR	
GPIO22_U0_RI		GPIO10_U1_RI	
GPIO4_U2_CTS		GPIO6_U3_CTS	
GPIO11_U2_DCD		GPIO15_U3_DCD	
GPIO12_U2_DTR		GPIO16_U3_DTR	
GPIO14_U2_RI		GPIO18_U3_RI	
GPIO1_U0_RTS		GPIO9_U1_DSR	
GPIO21_U0_DSR		GPIO3_U1_RTS	
GPIO5_U2_RTS		GPIO7_U3_RTS	
GPIO13_U2_DSR		GPIO17_U3_DSR	

SIM7080_RXD → uart0_RXD, SIM7080_TXD → uart0_TXD
GPS_RXD → uart1_RXD, GPS_TXD → uart1_TXD
RANGE_RXD → uart2_RXD, RANGE_TXD → uart2_TXD
VTX_RXD → uart3_RXD, VTX_TXD → uart3_TXD

JOY_UP	
JOY_LEFT	
JOY_DOWN	
JOY_RIGHT	
JOY_PUSH	
TCA_RST	
LEPTON_PWR_EN	
LEPTON_ENABLE	
LEPTON_RST	
RADIO_RST	
SIM7080_PWRKEY	
HARD_RESET	
TFT_RST	
◇TFT_VCC_IO	
◇TFT_VCC	

Fájl: SPI_to_UART.kicad_sch

Camera Connector

Display Connector

J7
Conn_01x15_Pin

1		
2		GND
3		
4		
5		GND
6		
7		
8		
9	TFT_RST	GND
10		
11	◇TFT_VCC_IO	
12	◇TFT_VCC	
13		
14	Sensors_3.3V	GND
15	R53 33ohm	

GND

SW1
RKJXM1015001

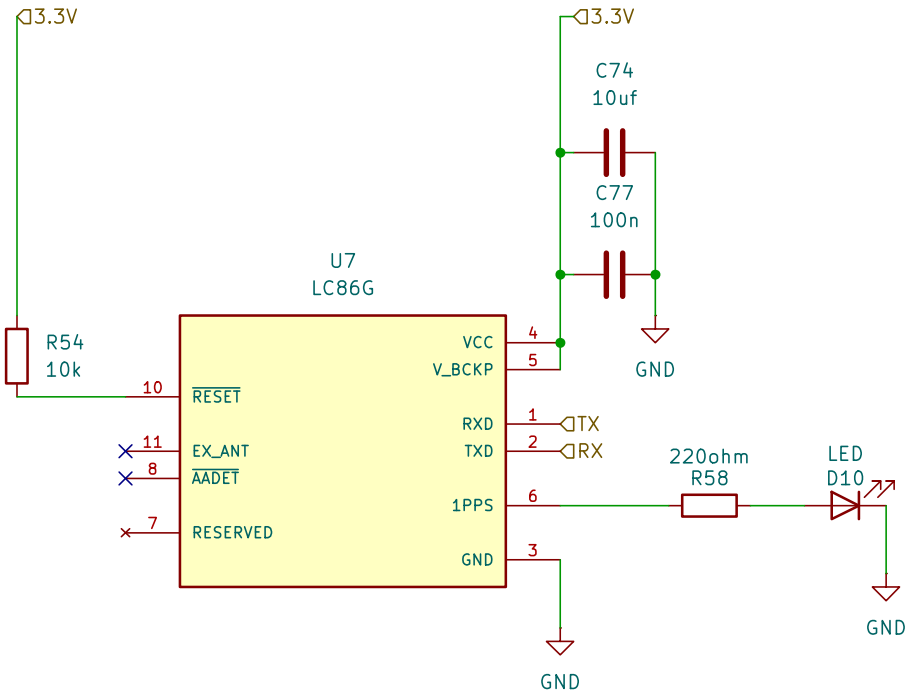
COM		A	JOY_UP
JOY_PUSH	PUSH	B	JOY_LEFT
GND		C	JOY_DOWN
		D	JOY_RIGHT

GND

Sheet: /Panel1_MCU/
File: Panel1_MCU.kicad_sch

Title:

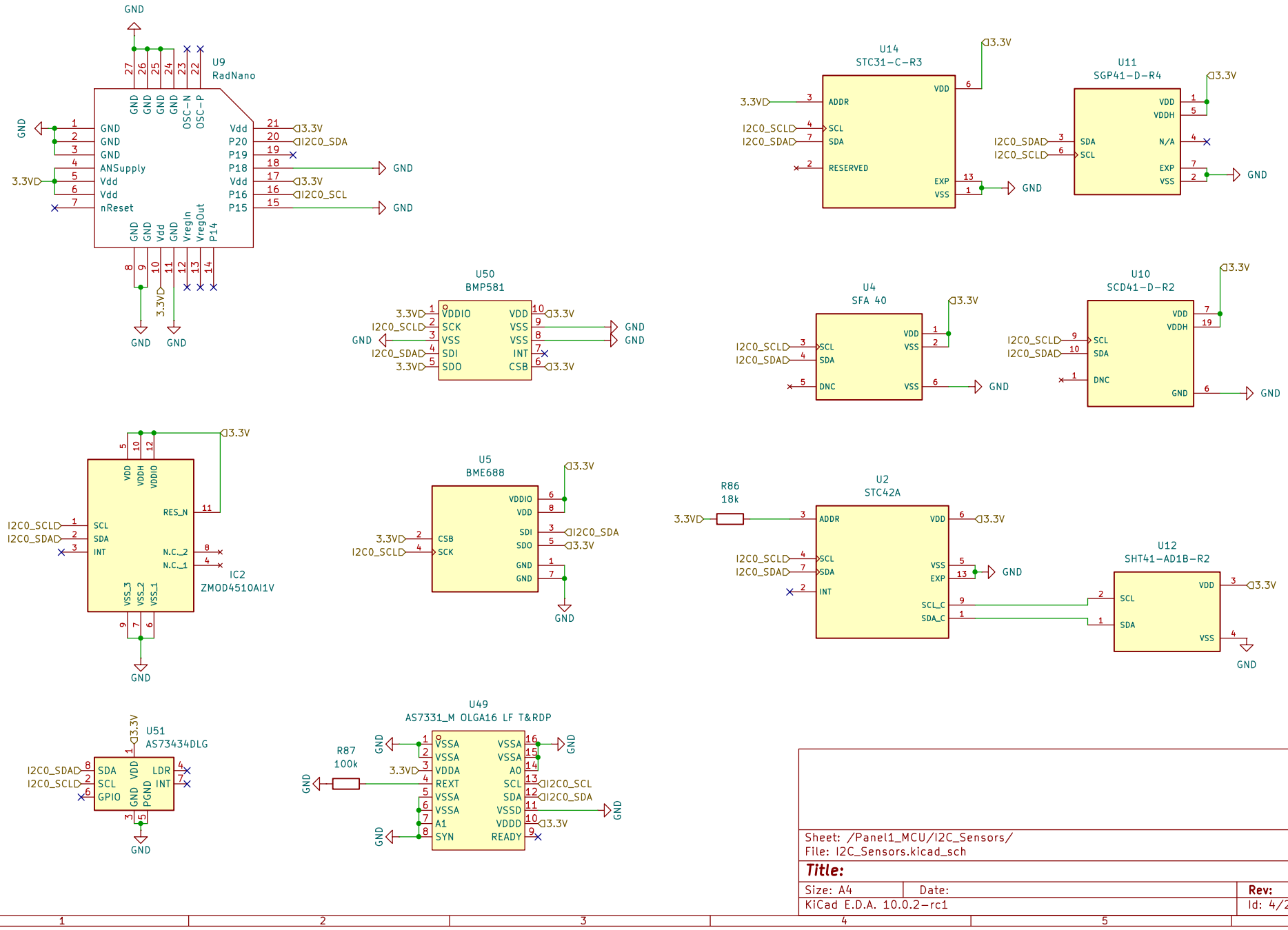
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.2-rc1		Id: 2/22

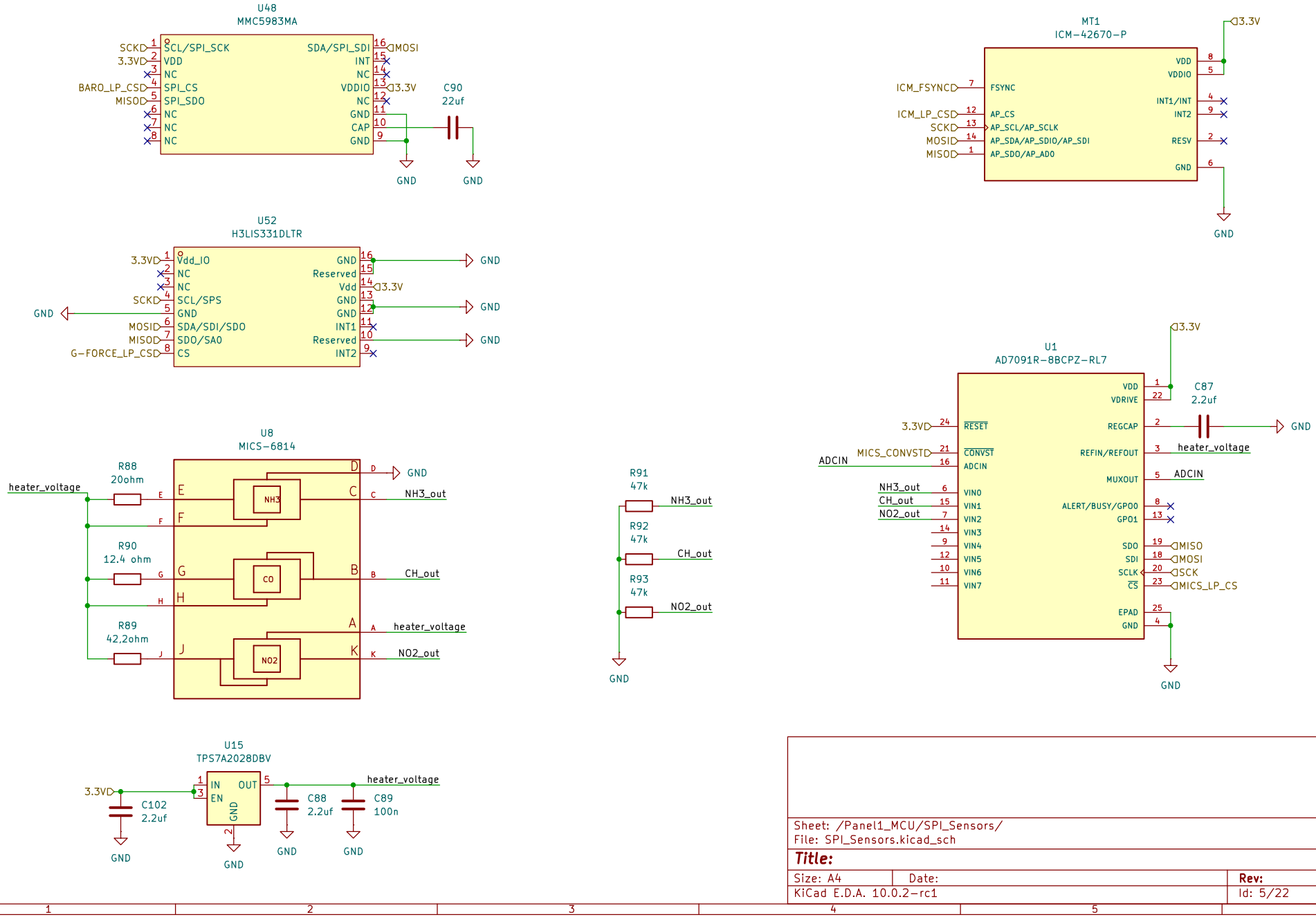


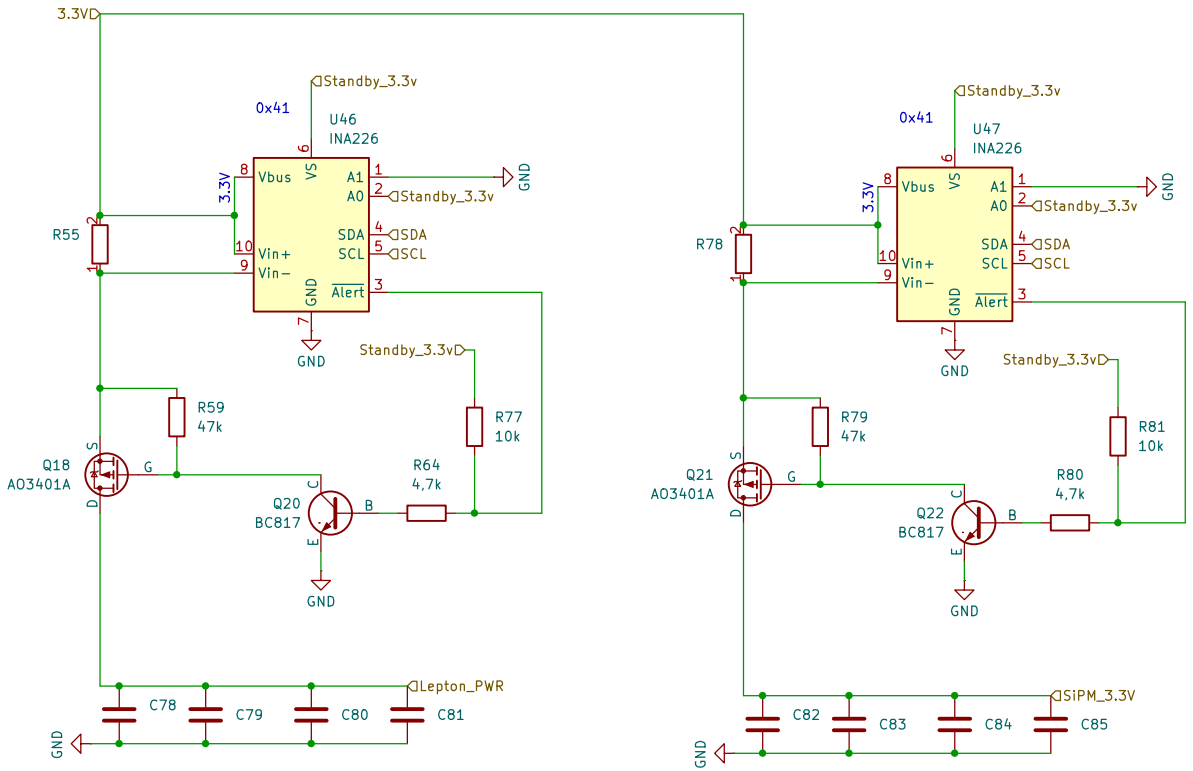
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File: GPS.kicad_sch

Title:

Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 3/22


U9
RadNano
GND
3.3V
I2C0_SDA
I2C0_SCL
U14
STC31-C-R3
ADDR
SCL
SDA
RESERVED
VDD
EXP
VSS
U11
SGP41-D-R4
VDDH
N/A
U50
BMP581
VDDIO
SCK
SDI
SDO
CSB
INT
VSS
U4
SFA 40
DNC
U10
SCD41-D-R2
U5
BME688
U2
STC42A
R86
18k
U12
SHT41-AD1B-R2
IC2
ZMOD4510AI1V
RES_N
N.C._2
N.C._1
U51
AS73434DLG
LDR
GPIO
U49
AS7331_M OLGA16 LF T&RDP
R87
100k
Sheet: /Panel1_MCU/I2C_Sensors/
File: I2C_Sensors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 4/22

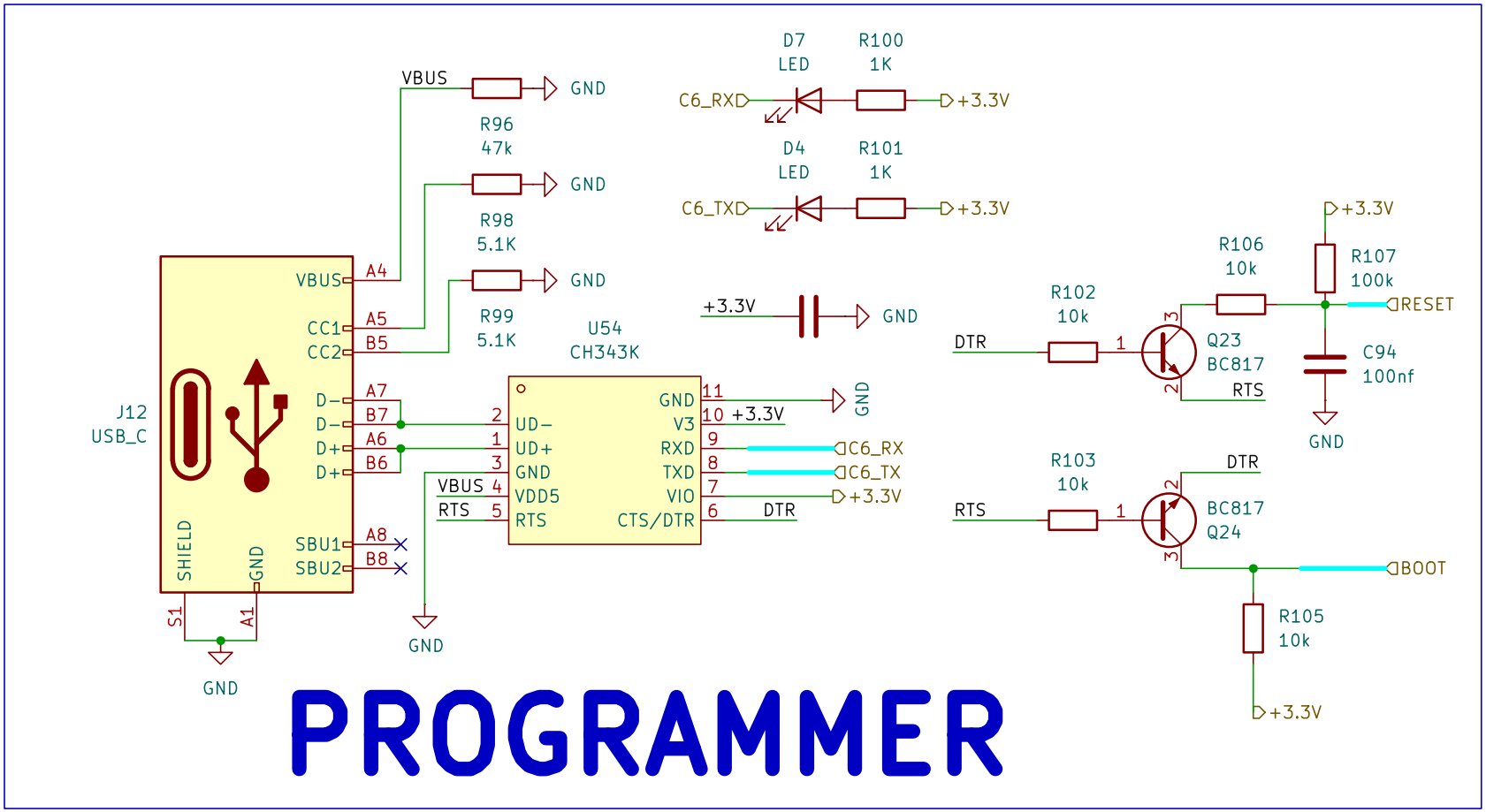

U48
MMC5983MA
SCK
3.3VD
BARO_LP_CSD
MISOD
MOSI
3.3V
C90
22uf
GND
U52
H3LIS331DLTR
3.3VD
SCKD
MOSID
MISOD
G-FORCE_LP_CSD
MT1
ICM-42670-P
ICM_FSYNCD
ICM_LP_CSD
SCKD
MOSID
MISOD
U8
MICS-6814
heater_voltage
R88
20ohm
R90
12.4 ohm
R89
42.2ohm
NH3_out
CH_out
NO2_out
R91
47k
R92
47k
R93
47k
U1
AD7091R-8BCPZ-RL7
3.3VD
MICS_CONVSTD
ADCIN
NH3_out
CH_out
NO2_out
C87
2.2uf
heater_voltage
ADCIN
MISO
MOSI
SCK
MICS_LP_CS
U15
TPS7A2028DBV
C102
2.2uf
C88
2.2uf
C89
100n
Sheet: /Panel1_MCU/SPI_Sensors/
File: SPI_Sensors.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 5/22



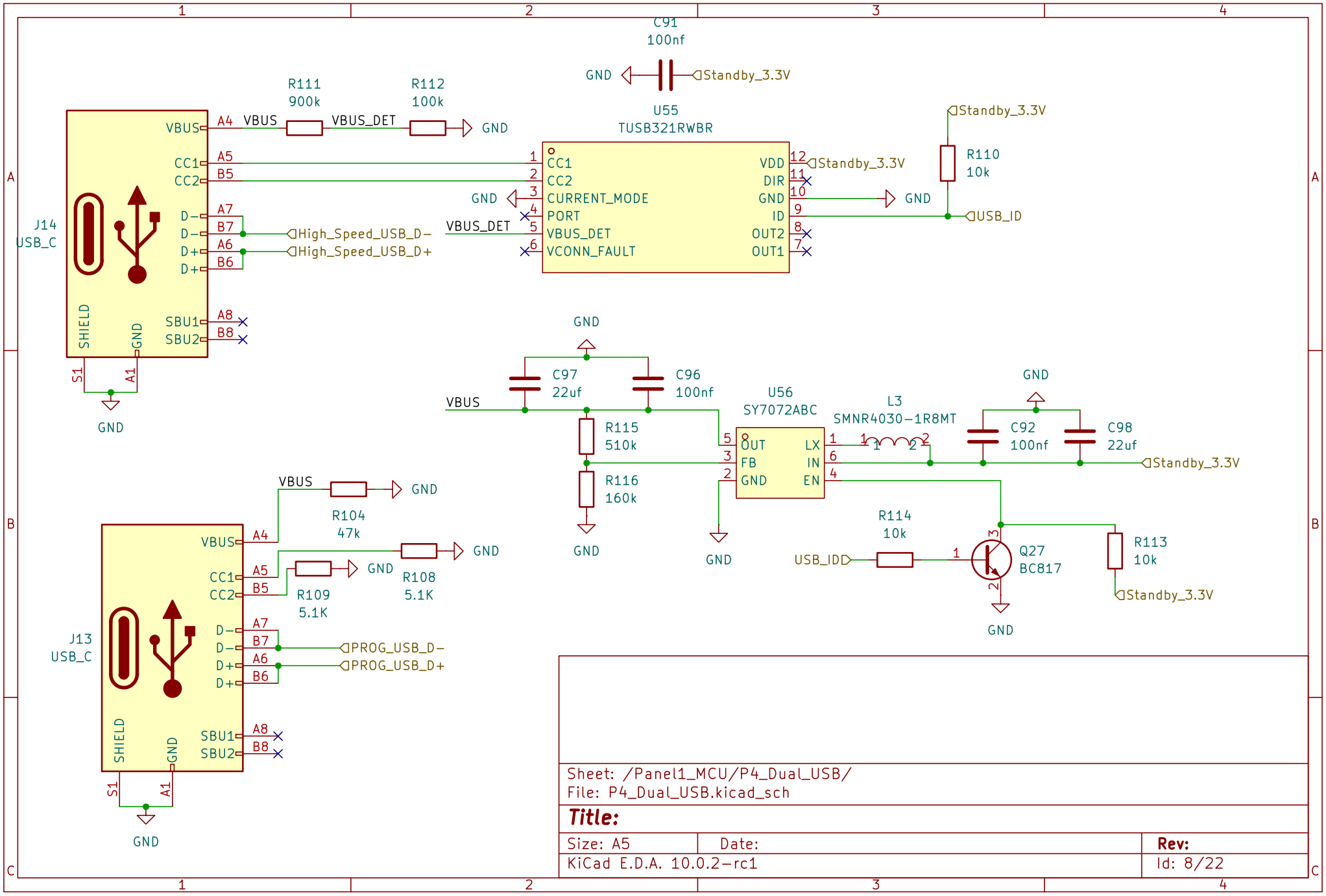
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File: Lepton_SiPM_Power.kicad_sch

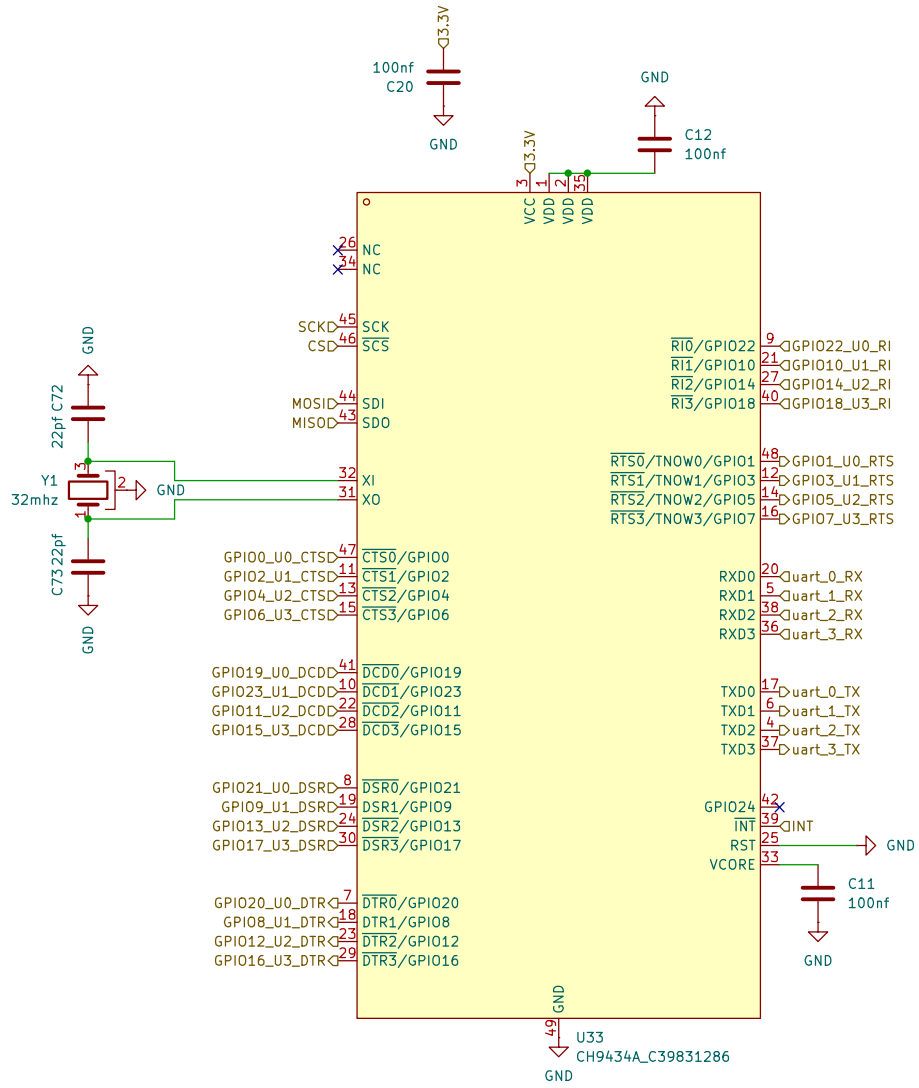
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 6/22



Sheet: /Panel1_MCU/C6_Programmer/		
File: C6_Programmer.kicad_sch		
Title:		
Size: A5	Date:	**Rev:**
KiCad E.D.A. 10.0.2−rc1		Id: 7/22

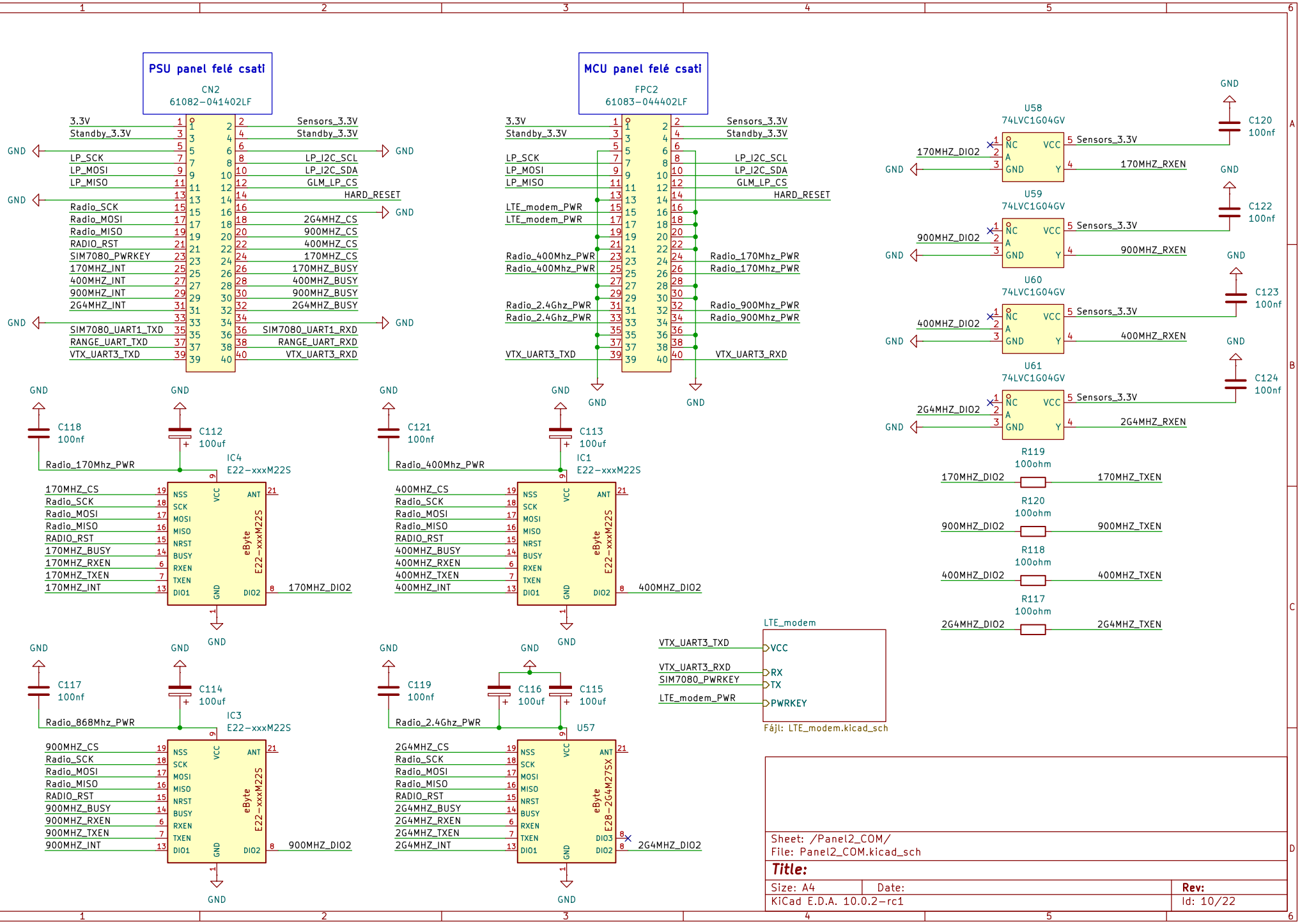

C91
100nf
GND
Standby_3.3V
R111
900k
R112
100k
U55
TUSB321RWBR
VBUS
VBUS_DET
GND
J14
USB_C
CC1
CC2
D−
D+
SHIELD
SBU1
SBU2
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
High_Speed_USB_D−
High_Speed_USB_D+
CURRENT_MODE
PORT
VCONN_FAULT
VDD
DIR
ID
OUT2
OUT1
R110
10k
USB_ID
C97
22uf
C96
100nf
R115
510k
R116
160k
U56
SY7072ABC
OUT
FB
EN
LX
IN
L3
SMNR4030−1R8MT
C92
100nf
C98
22uf
R104
47k
R108
5.1K
R109
5.1K
J13
USB_C
PROG_USB_D−
PROG_USB_D+
R114
10k
Q27
BC817
R113
10k
Sheet: /Panel1_MCU/P4_Dual_USB/
File: P4_Dual_USB.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2−rc1
Id: 8/22

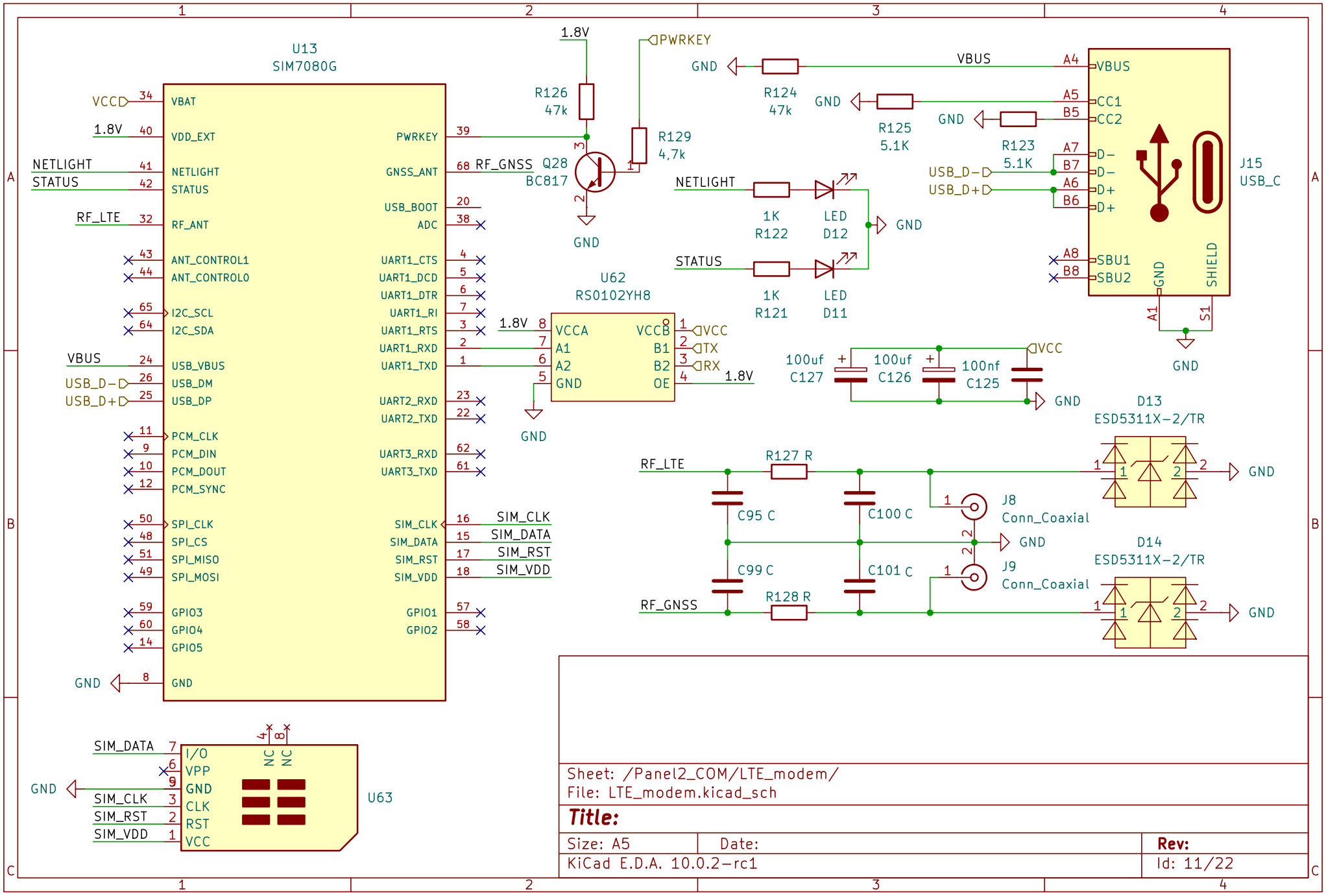


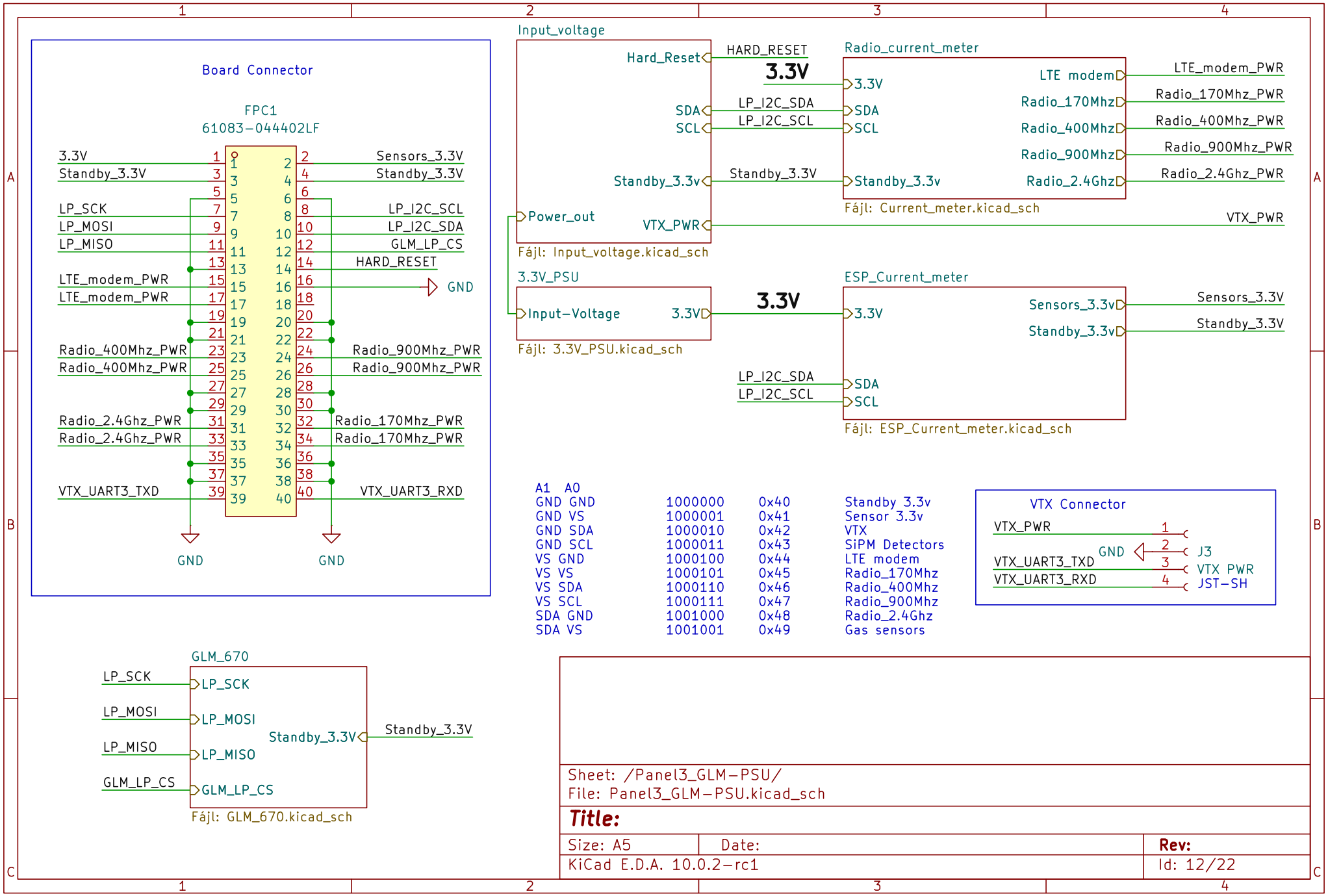
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File: SPI_to_UART.kicad_sch

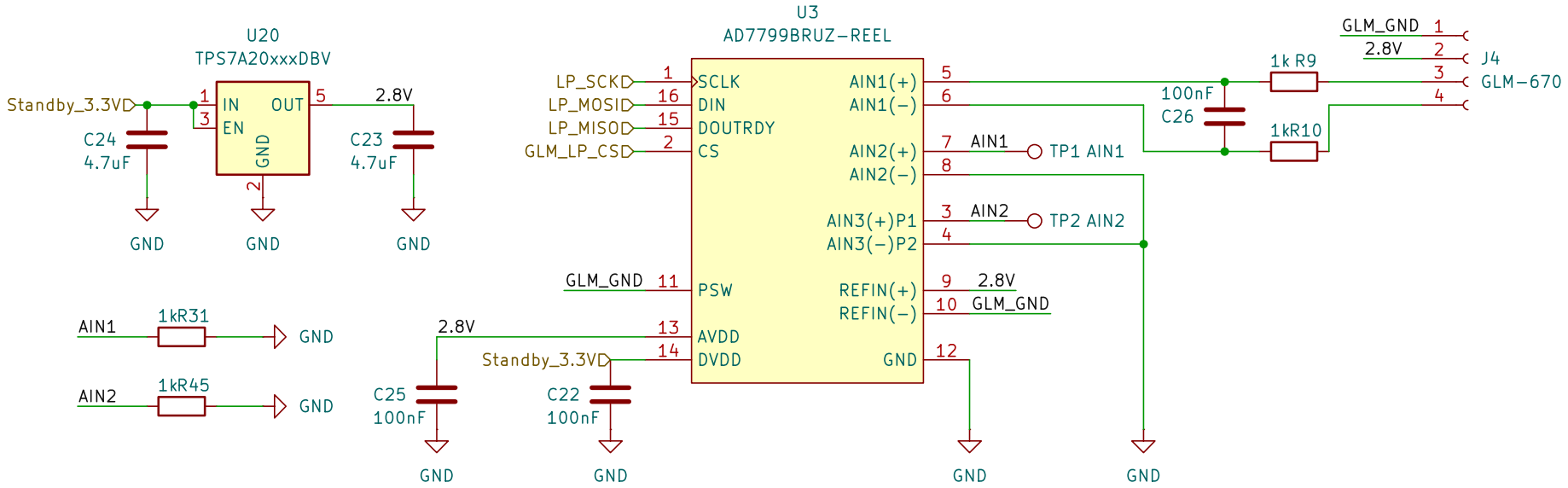
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.2-rc1		Id: 9/22

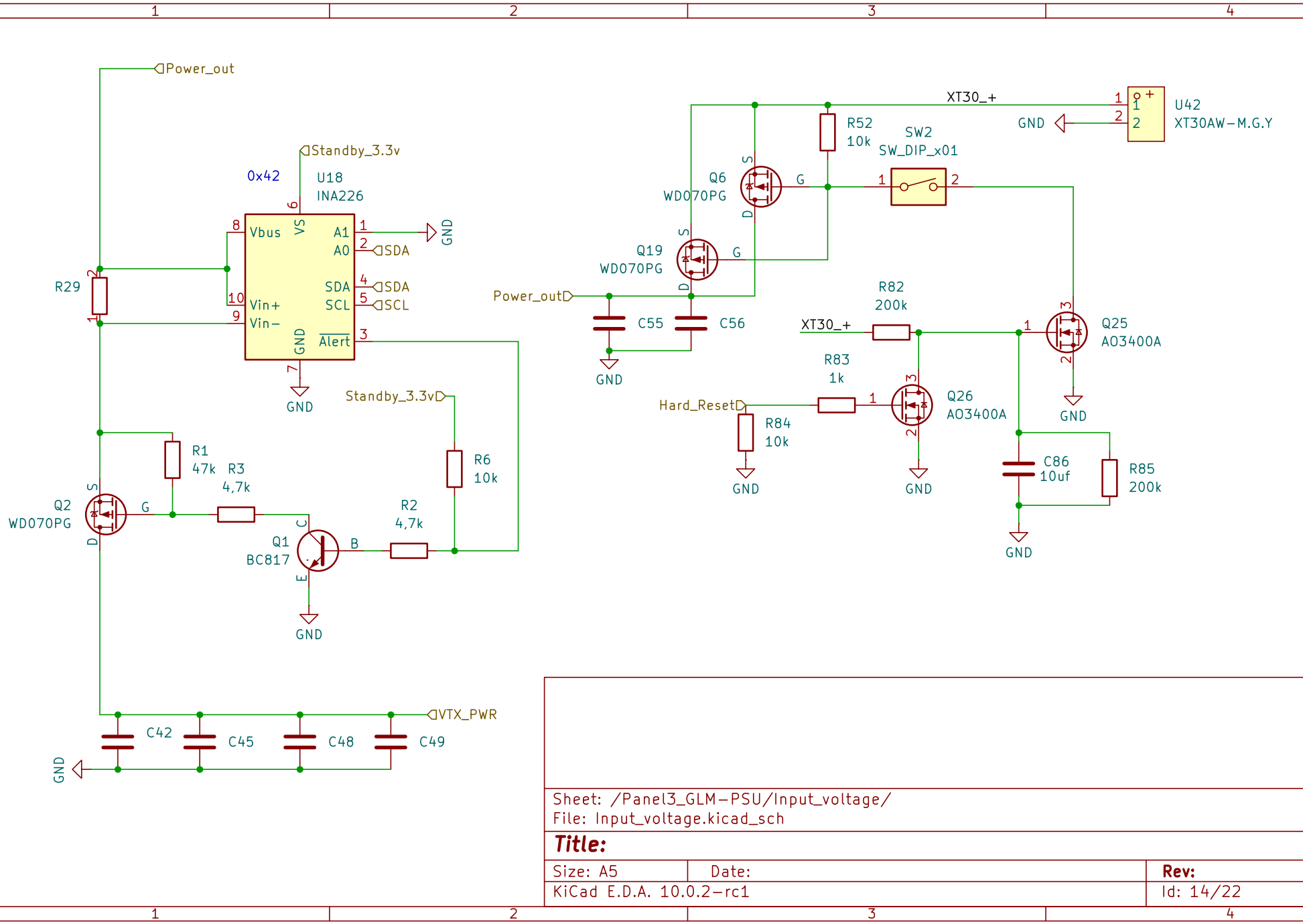

PSU panel felé csati
CN2
61082-041402LF
MCU panel felé csati
FPC2
61083-044402LF
3.3V
Standby_3.3V
Sensors_3.3V
Standby_3.3V
GND
LP_SCK
LP_MOSI
LP_MISO
LP_I2C_SCL
LP_I2C_SDA
GLM_LP_CS
HARD_RESET
Radio_SCK
Radio_MOSI
Radio_MISO
RADIO_RST
SIM7080_PWRKEY
170MHZ_INT
400MHZ_INT
900MHZ_INT
2G4MHZ_INT
2G4MHZ_CS
900MHZ_CS
400MHZ_CS
170MHZ_CS
170MHZ_BUSY
400MHZ_BUSY
900MHZ_BUSY
2G4MHZ_BUSY
SIM7080_UART1_TXD
SIM7080_UART1_RXD
RANGE_UART_TXD
RANGE_UART_RXD
VTX_UART3_TXD
VTX_UART3_RXD
LTE_modem_PWR
Radio_400Mhz_PWR
Radio_170Mhz_PWR
Radio_2.4Ghz_PWR
Radio_900Mhz_PWR
U58
74LVC1G04GV
U59
U60
U61
NC
VCC
A
GND
Y
170MHZ_DIO2
900MHZ_DIO2
400MHZ_DIO2
2G4MHZ_DIO2
170MHZ_RXEN
900MHZ_RXEN
400MHZ_RXEN
2G4MHZ_RXEN
C120
C122
C123
C124
100nf
C118
C112
C121
C113
C117
C114
C119
C116
C115
100uf
Radio_170Mhz_PWR
Radio_400Mhz_PWR
Radio_868Mhz_PWR
Radio_2.4Ghz_PWR
IC4
IC1
IC3
U57
E22-xxxM22S
E28-2G4M27SX
eByte
NSS
SCK
MOSI
MISO
NRST
BUSY
RXEN
TXEN
DIO1
DIO2
DIO3
ANT
VCC
170MHZ_CS
400MHZ_CS
900MHZ_CS
2G4MHZ_CS
170MHZ_TXEN
400MHZ_TXEN
900MHZ_TXEN
2G4MHZ_TXEN
R119
R120
R118
R117
100ohm
170MHZ_TXEN
900MHZ_TXEN
400MHZ_TXEN
2G4MHZ_TXEN
LTE_modem
RX
TX
PWRKEY
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File: Panel2_COM.kicad_sch
Title:
Size: A4
Date:
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U13
SIM7080G
VCC
1.8V
NETLIGHT
STATUS
RF_LTE
VBUS
USB_D-
USB_D+
VBAT
VDD_EXT
NETLIGHT
STATUS
RF_ANT
ANT_CONTROL1
ANT_CONTROL0
I2C_SCL
I2C_SDA
USB_VBUS
USB_DM
USB_DP
PCM_CLK
PCM_DIN
PCM_DOUT
PCM_SYNC
SPI_CLK
SPI_CS
SPI_MISO
SPI_MOSI
GPIO3
GPIO4
GPIO5
GND
PWRKEY
GNSS_ANT
USB_BOOT
ADC
UART1_CTS
UART1_DCD
UART1_DTR
UART1_RI
UART1_RTS
UART1_RXD
UART1_TXD
UART2_RXD
UART2_TXD
UART3_RXD
UART3_TXD
SIM_CLK
SIM_DATA
SIM_RST
SIM_VDD
GPIO1
GPIO2
R126
47k
R129
4,7k
Q28
BC817
RF_GNSS
U62
RS0102YH8
VCCA
A1
A2
GND
VCCB
B1
B2
OE
VCC
TX
RX
R124
47k
R125
5.1K
R123
5.1K
J15
USB_C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
R122
1K
LED
D12
R121
1K
LED
D11
C127
100uf
C126
100uf
C125
100nf
D13
ESD5311X-2/TR
D14
ESD5311X-2/TR
R127 R
R128 R
C95 C
C100 C
C99 C
C101 C
J8
Conn_Coaxial
J9
Conn_Coaxial
U63
I/O
VPP
GND
CLK
RST
VCC
NC
NC
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File: LTE_modem.kicad_sch
Title:
Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
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Board Connector
FPC1
61083-044402LF
3.3V
Standby_3.3V
LP_SCK
LP_MOSI
LP_MISO
LTE_modem_PWR
LTE_modem_PWR
Radio_400Mhz_PWR
Radio_400Mhz_PWR
Radio_2.4Ghz_PWR
Radio_2.4Ghz_PWR
VTX_UART3_TXD
Sensors_3.3V
Standby_3.3V
LP_I2C_SCL
LP_I2C_SDA
GLM_LP_CS
HARD_RESET
GND
Radio_900Mhz_PWR
Radio_900Mhz_PWR
Radio_170Mhz_PWR
Radio_170Mhz_PWR
VTX_UART3_RXD
GND
GND
Input_voltage
Hard_Reset
HARD_RESET
3.3V
SDA
SCL
LP_I2C_SDA
LP_I2C_SCL
Standby_3.3v
Standby_3.3V
Power_out
VTX_PWR
Fájl: Input_voltage.kicad_sch
Radio_current_meter
3.3V
SDA
SCL
Standby_3.3v
LTE modem
Radio_170Mhz
Radio_400Mhz
Radio_900Mhz
Radio_2.4Ghz
LTE_modem_PWR
Radio_170Mhz_PWR
Radio_400Mhz_PWR
Radio_900Mhz_PWR
Radio_2.4Ghz_PWR
VTX_PWR
Fájl: Current_meter.kicad_sch
3.3V_PSU
Input-Voltage
3.3V
3.3V
Fájl: 3.3V_PSU.kicad_sch
ESP_Current_meter
3.3V
LP_I2C_SDA
LP_I2C_SCL
SDA
SCL
Sensors_3.3v
Standby_3.3v
Sensors_3.3V
Standby_3.3V
Fájl: ESP_Current_meter.kicad_sch
A1 A0
GND GND 1000000 0x40 Standby 3.3v
GND VS 1000001 0x41 Sensor 3.3v
GND SDA 1000010 0x42 VTX
GND SCL 1000011 0x43 SiPM Detectors
VS GND 1000100 0x44 LTE modem
VS VS 1000101 0x45 Radio_170Mhz
VS SDA 1000110 0x46 Radio_400Mhz
VS SCL 1000111 0x47 Radio_900Mhz
SDA GND 1001000 0x48 Radio_2.4Ghz
SDA VS 1001001 0x49 Gas sensors
VTX Connector
VTX_PWR 1
GND 2 J3
VTX_UART3_TXD 3 VTX PWR
VTX_UART3_RXD 4 JST-SH
GLM_670
LP_SCK
LP_MOSI
LP_MISO
GLM_LP_CS
LP_SCK
LP_MOSI
LP_MISO
GLM_LP_CS
Standby_3.3V
Standby_3.3V
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Size: A5
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 12/22

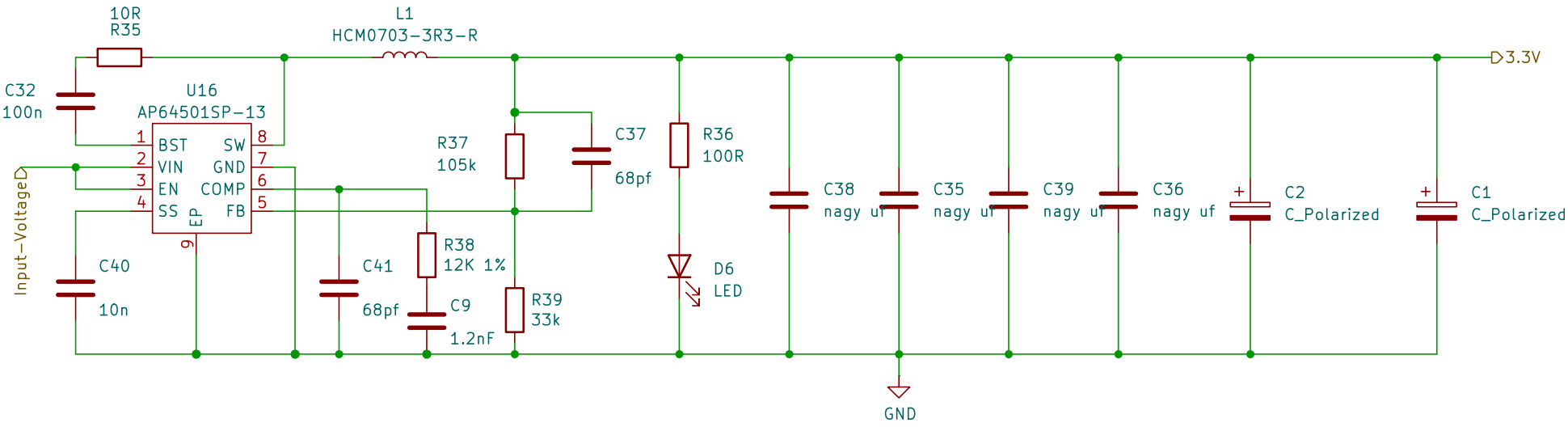


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KiCad E.D.A. 10.0.2-rc1		Id: 13/22

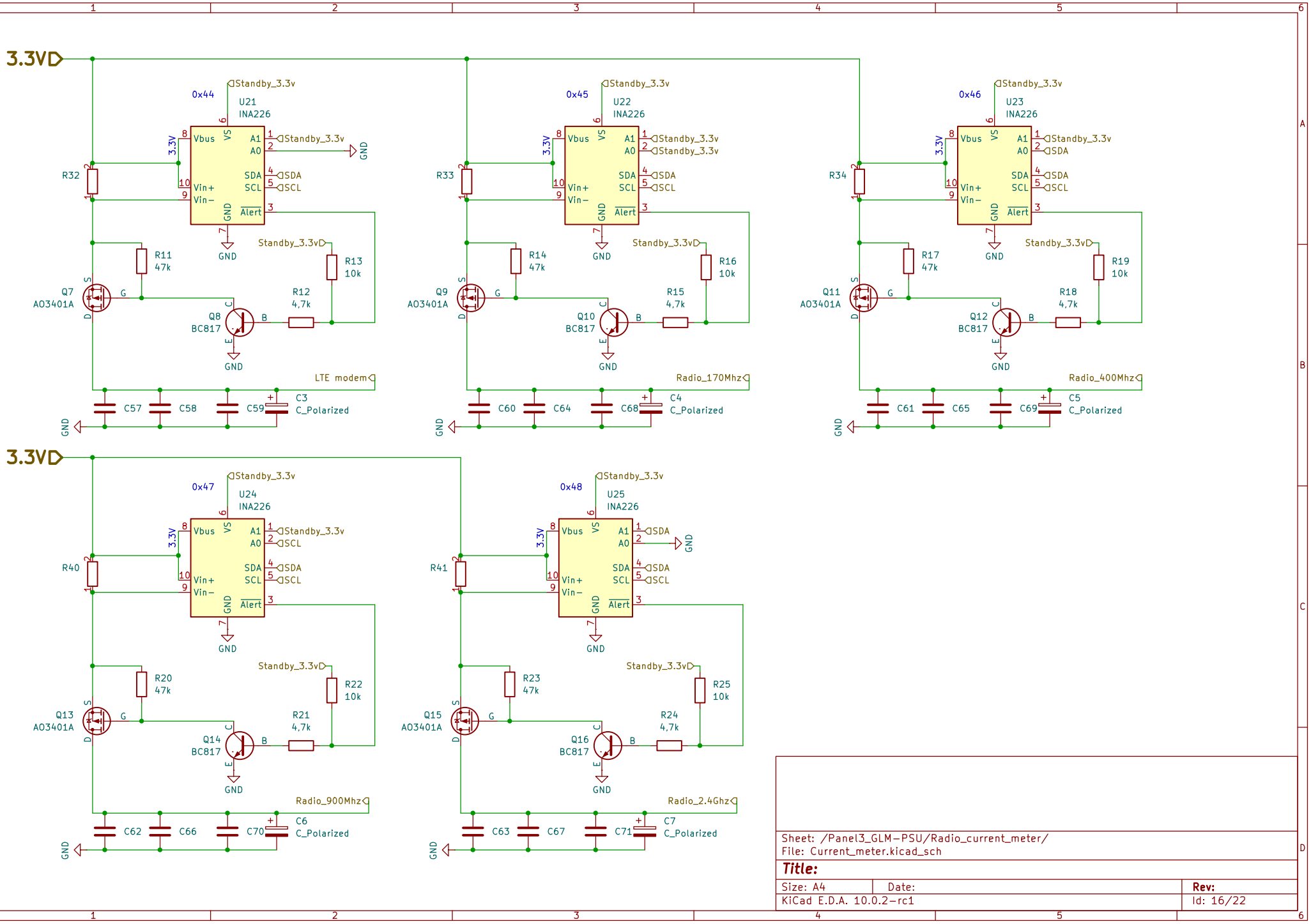


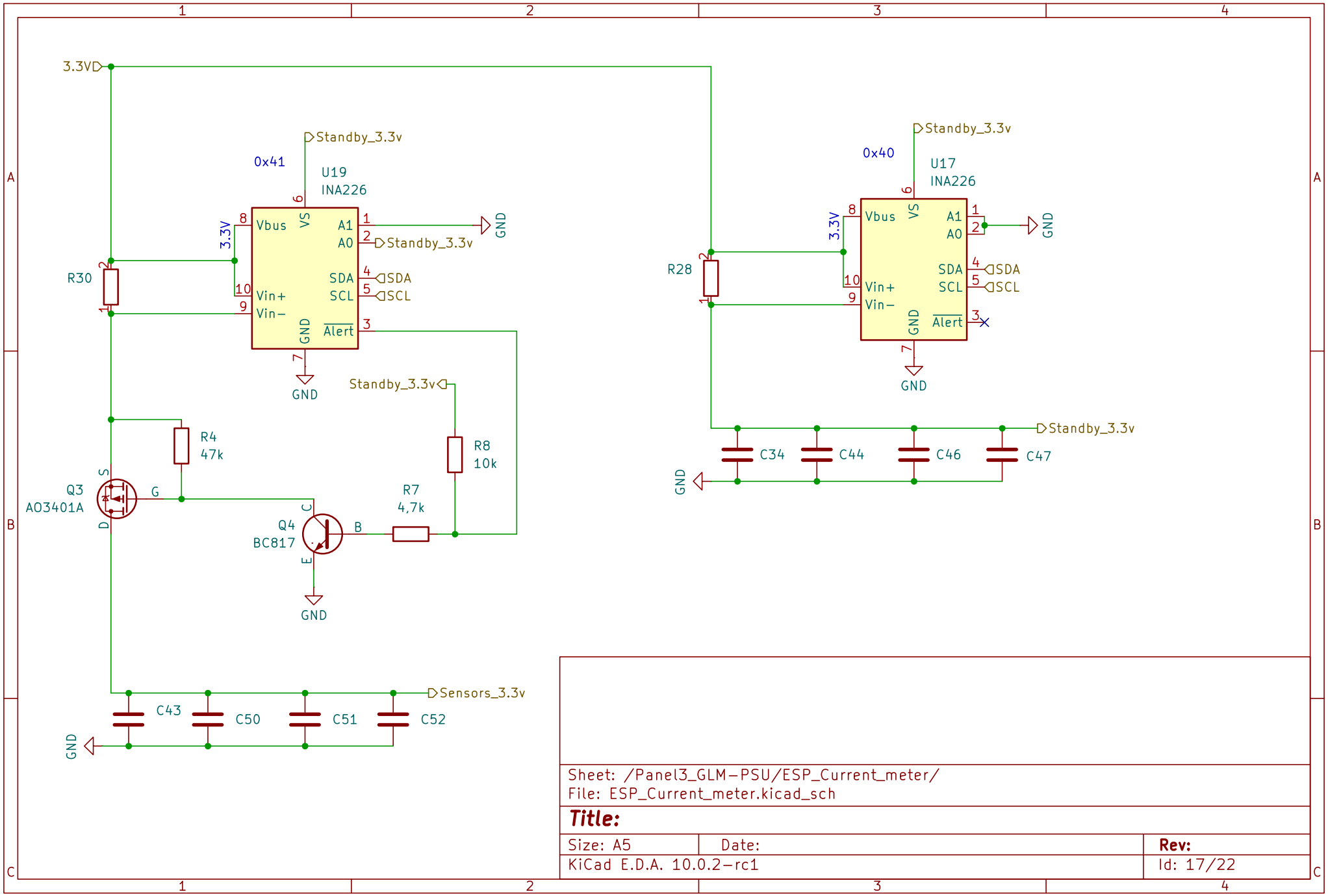
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Size: A5	Date:	**Rev:**
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3.3v @ 5A PSU



Sheet: /Panel3_GLM-PSU/3.3V_PSU/		
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Size: A5	Date:	**Rev:**
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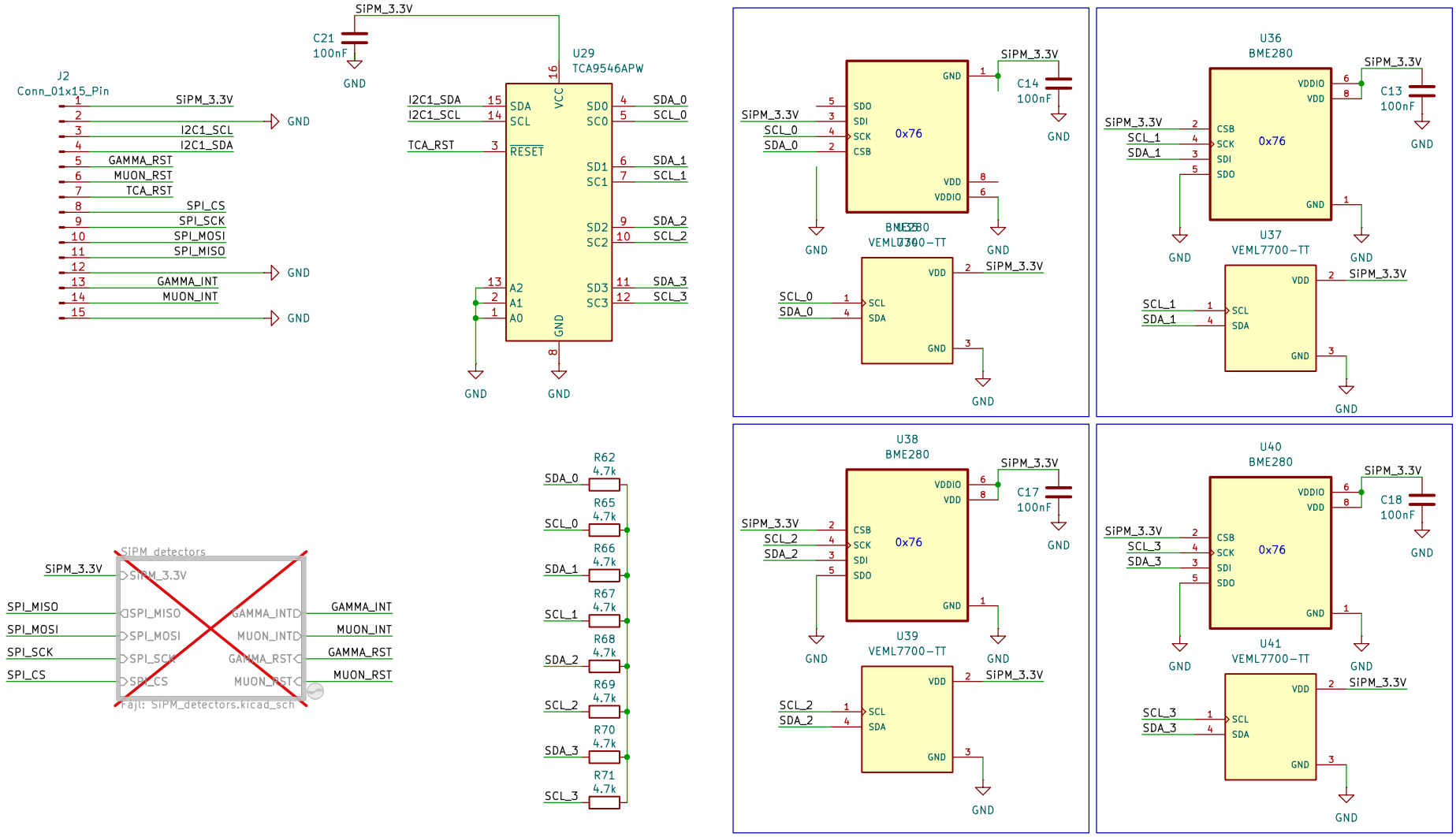

3.3VD
0x44
U21
INA226
Standby_3.3v
R32
R11
47k
R13
10k
R12
4,7k
Q7
AO3401A
Q8
BC817
LTE modem
C57
C58
C59
C3
C_Polarized
0x45
U22
INA226
R33
R14
47k
R16
10k
R15
4,7k
Q9
AO3401A
Q10
BC817
Radio_170Mhz
C60
C64
C68
C4
C_Polarized
0x46
U23
INA226
SDA
R34
R17
47k
R19
10k
R18
4,7k
Q11
AO3401A
Q12
BC817
Radio_400Mhz
C61
C65
C69
C5
C_Polarized
3.3VD
0x47
U24
INA226
SCL
R40
R20
47k
R22
10k
R21
4,7k
Q13
AO3401A
Q14
BC817
Radio_900Mhz
C62
C66
C70
C6
C_Polarized
0x48
U25
INA226
R41
R23
47k
R25
10k
R24
4,7k
Q15
AO3401A
Q16
BC817
Radio_2.4Ghz
C63
C67
C71
C7
C_Polarized
GND
Vbus
Vin+
Vin−
SDA
SCL
Alert
A1
A0
VS
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Size: A4
Date:
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KiCad E.D.A. 10.0.2-rc1
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File: ESP_Current_meter.kicad_sch

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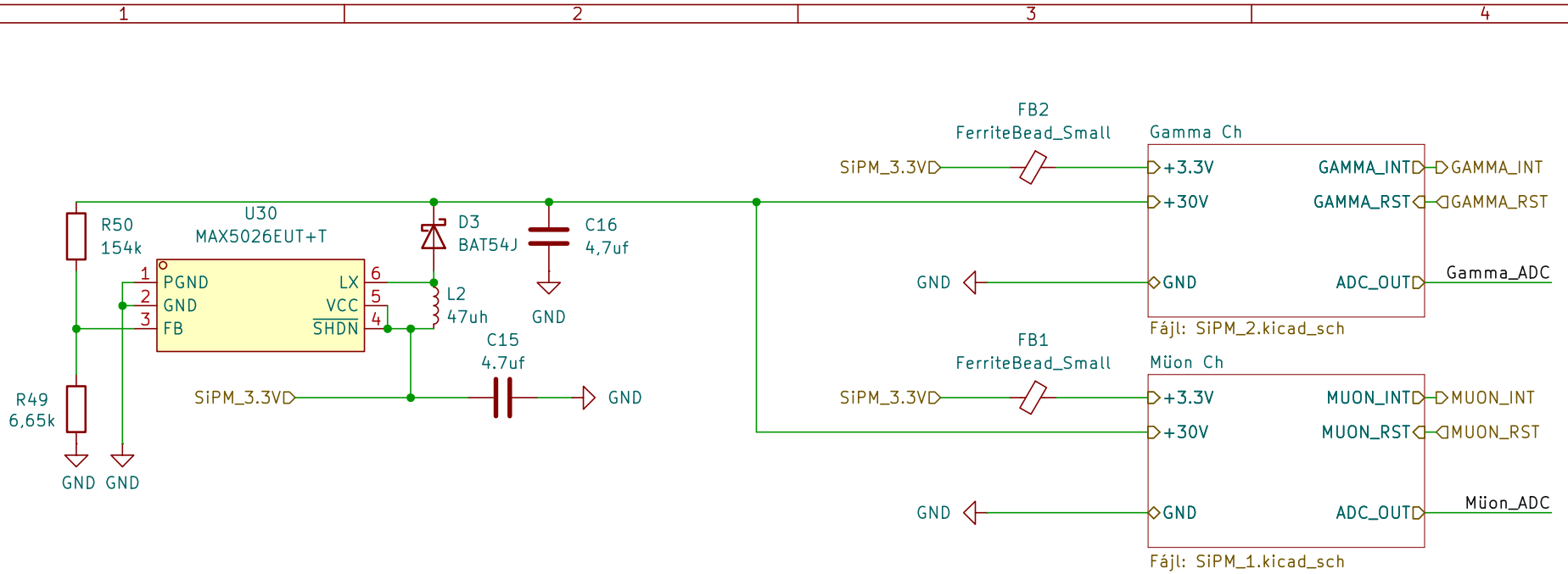
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KiCad E.D.A. 10.0.2-rc1		Id: 17/22



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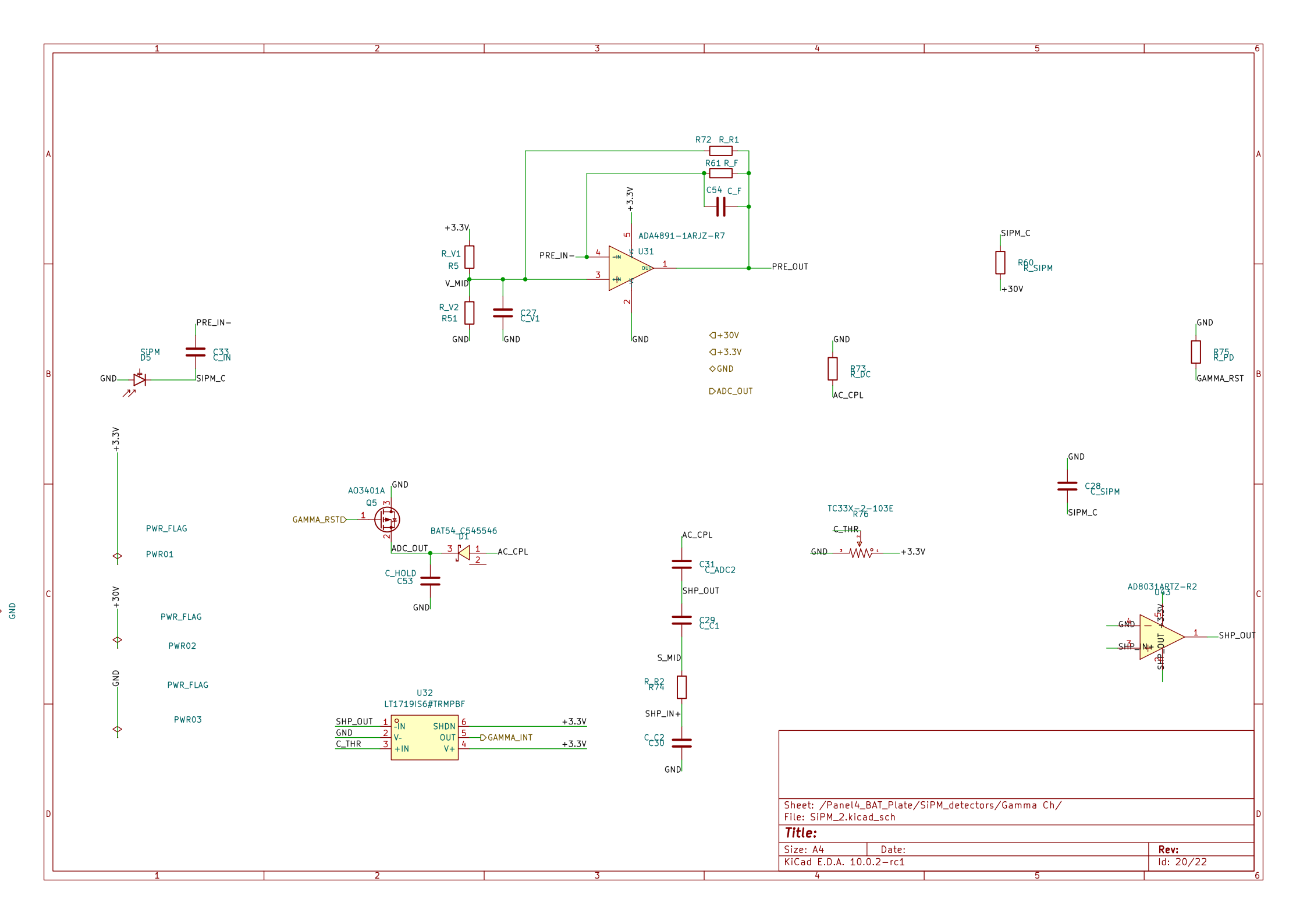
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Sheet: /Panel4_BAT_Plate/SiPM_detectors/
File: SiPM_detectors.kicad_sch

Title:

Size: A5	Date:	**Rev:**
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R72 R_R1
R61 R_F
C54 C_F
+3.3V
R_V1
R5
V_MID
R_V2
R51
GND
C27
C_V1
ADA4891-1ARJZ-R7
U31
PRE_IN-
PRE_OUT
SIPM_C
R60
R_SIPM
+30V
GND
R75
R_PD
GAMMA_RST
SIPM
D5
C33
C_IN
+30V
+3.3V
GND
ADC_OUT
R73
R_DC
AC_CPL
PWR_FLAG
PWR01
PWR02
PWR03
AO3401A
Q5
GAMMA_RST
BAT54_C545546
D1
ADC_OUT
AC_CPL
C_HOLD
C53
C28
C_SIPM
TC33X-2-103E
R76
C_THR
C31
C_ADC2
SHP_OUT
C29
C_C1
S_MID
R_R2
R74
SHP_IN+
C_C2
C30
AD8031ARTZ-R2
U43
U32
LT1719IS6#TRMPBF
SHP_OUT
GND
C_THR
-IN
V-
+IN
SHDN
OUT
V+
GAMMA_INT
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File: SiPM_2.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 20/22

+30V

+3.3V

GND

ADC_OUT

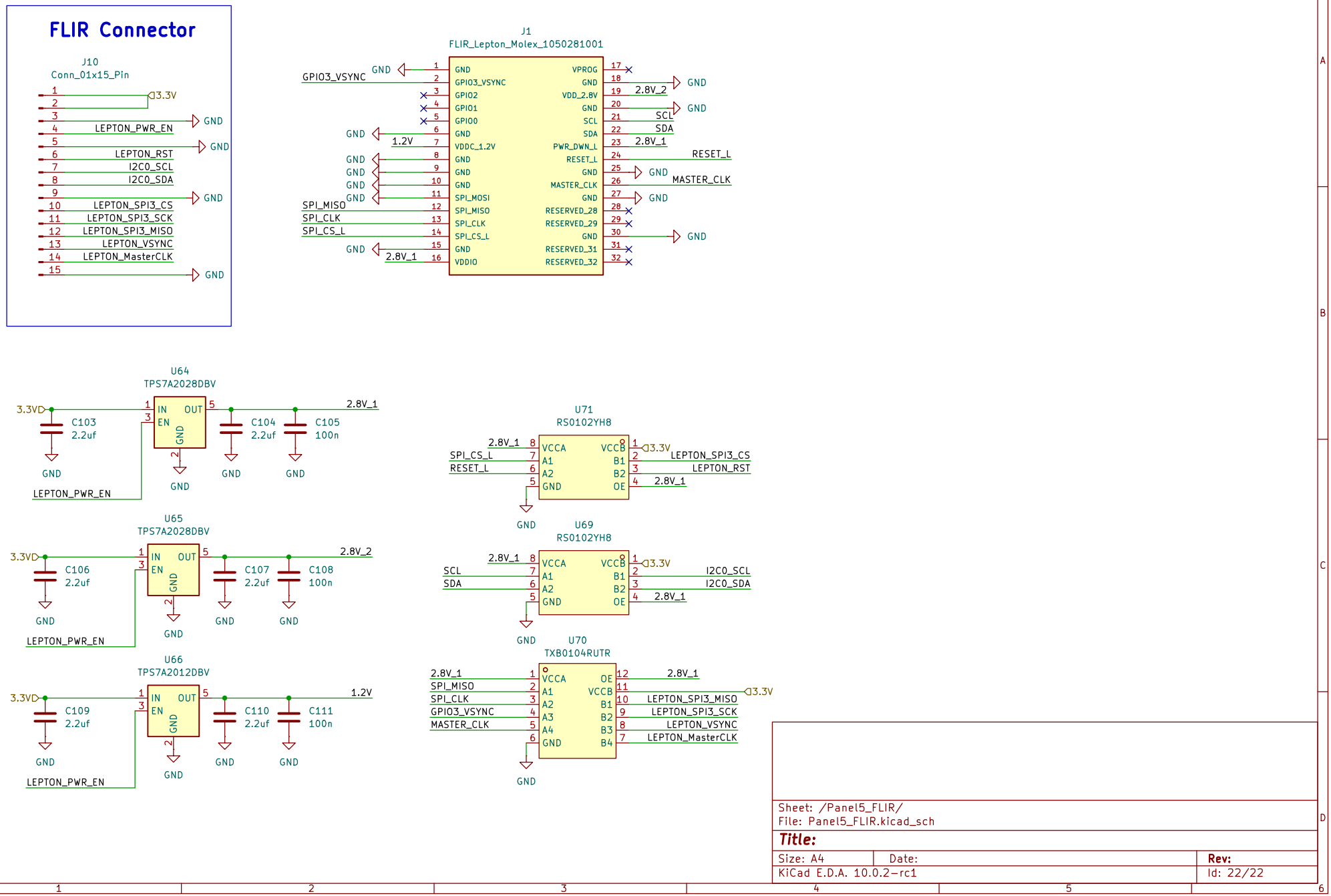
MUON_INT

MUON_RST

Sheet: /Panel4_BAT_Plate/SiPM_detectors/Müon Ch/
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Title:

Size: A4	Date:	**Rev:**
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FLIR Connector
J10
Conn_01x15_Pin
3.3V
GND
LEPTON_PWR_EN
LEPTON_RST
I2C0_SCL
I2C0_SDA
LEPTON_SPI3_CS
LEPTON_SPI3_SCK
LEPTON_SPI3_MISO
LEPTON_VSYNC
LEPTON_MasterCLK
J1
FLIR_Lepton_Molex_1050281001
GPIO3_VSYNC
SPI_MISO
SPI_CLK
SPI_CS_L
1.2V
2.8V_1
2.8V_2
SCL
SDA
RESET_L
MASTER_CLK
U64
TPS7A2028DBV
C103
2.2uf
C104
2.2uf
C105
100n
U65
TPS7A2028DBV
C106
2.2uf
C107
2.2uf
C108
100n
U66
TPS7A2012DBV
C109
2.2uf
C110
2.2uf
C111
100n
U71
RS0102YH8
U69
RS0102YH8
U70
TXB0104RUTR
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Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.2-rc1
Id: 22/22